

A graphic with a black background. On the left, a green, pill-shaped microorganism with cilia and a smaller green sphere are connected by a white wavy line. On the right, a purple, spherical virus-like particle with spikes is connected to the text by a white wavy line. The text "CONTAGION CHRONICLES" is in large, bold, yellow-green capital letters, and "THE COVID-19 JOURNEY" is in smaller, white capital letters below it.

Caused by the new coronavirus SARS-CoV-2. COVID-19. The main ways that this virus spreads through the respiratory, when COVID-19 presents clinically, symptoms might vary from none at all to a serious respiratory condition that results in death[1].

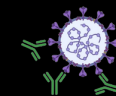
The immune system serves as the body's line of defense against viruses such as SARS-CoV-2. Antibodies are proteins made by B cells (a subset of white blood cells) that play a major role in this response[2]. Pathogen surfaces contain distinct molecules called antigens, which are recognized and bound to by antibodies[3].

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HOW ANTIBODIES WORK

1. Neutralization

By attaching to the surface proteins of viruses, antibodies can neutralize them and stop them from entering and infecting cells[4].



2.Cpsonization

In this process, antibodies label pathogens so that other immune cells, like neutrophils and macrophages, can destroy them[4].



3. Complement Activation

The complement system is a set of proteins that help destroy pathogens and can be activated by antibodies[4].



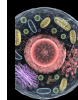
4. Antibody-Dependent Cellular Cytotoxicity (ADCC)

Through this mechanism, immune cells are drawn to infected cells by antibodies, which then kill the infected cells[4].



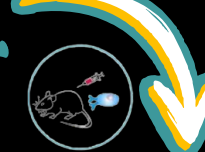
MONOCLONAL ANTIBODIES! A TARGET APPROACH

Monoclonal antibodies (mAbs) are laboratory-created molecules designed to function as substitute antibodies, restoring, enhancing, or mimicking the immune system's attack on infections[5].



Immunization

The target antigen is injected into a mouse or other suitable host[6].



MONOCLONAL ANTIBODY PRODUCTION

Extraction

B cells that produce antibodies against the antigen are retrieved from the host[7].

Selection and

Screening
Hybridoma cells are screened to determine which produce the desired antibodies[9].

Fusion

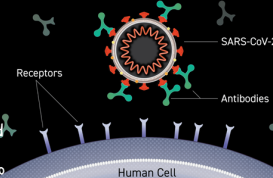
These B cells are combined with myeloma cells, which are cancer cells capable of endless division, to form hybridoma cells, which are capable of perpetual antibody production[8].



Monoclonal

Antibodies Against Covid-19

These antibodies typically target the spike protein, which allows the virus to enter human cells. Monoclonal antibodies, by binding to the spike protein, can prevent the virus from adhering to and entering cells, effectively neutralising its capacity to infect[10].



Mechanism of Action

These antibodies can enlist additional immune system components to aid in the removal of the virus and infected cells by binding to the spike protein of SARS-CoV-2. This prevents the virus from adhering to the ACE2 receptors on human cells, which is the first stage in viral entry[11].

References

1. Crotti M, Cocca M, Terranova AY, Wang C-W, Bergamini S. The COVID-19 pandemic: Critical Reviews in Clinical Laboratory Sciences. *Crit Rev Clin Lab Sci*. 2020;67(1):1–18.
2. Janczewska R, Janczewski EM, Jacobyck M Understanding Antibody Testing for COVID-19. *J Immunol Methods*. 2020;473(1):125741–81.
3. Nogueira M, Cellario D, Daouk AO, Constantin C, Filippini T, Vincenzi M et al. Back to basics in COVID-19: Antigens and antibodies—Completing the puzzle. *Journal of Cellular and Molecular Medicine*. 2020;24(10):2453–65.
4. Kulkarni AB, Chakrabarti SK, Ghosh A, Dasgupta S, Ghoshal S, Ghoshal S et al. Molecular mechanisms behind combination therapy. *Current Opinion in Microbiology*. 2020;50 Oct 57:44–50.
5. Bhatia V, Garg P, Kumar S, Singh M, Ma A, Pickles R, Le SX. Challenges and opportunities for antiviral monoclonal antibodies as COVID-19 therapy. *Advanced Drug Delivery Reviews*. 2020;167 Dec 100:1–17.
6. Wileman C. [1950]. Monoclonal antibodies. *Scientific American*, 243(4), 66–76. <http://www.jstor.org/stable/24966455>.
7. Hackett JR, Fooks PJ, Smith RT, Brown IJ. Monoclonal antibody testing: A Overview of Neutralizing Antibodies and Their Potential in COVID-19 Vaccines. *2020 Nov 25*;12(1):376.
8. Deeb P, Molis MW, Selby LR, McKinnon MA. An update to monoclonal antibodies as therapeutic option for COVID-19. *Bioassay and Health*. 2021 Apr 15:1–15.
9. Deeb P, Molis MW, C. Truchiani K, McKinnon MA, Ma A, Pickles R, Le SX. Challenges and opportunities for antiviral monoclonal antibodies as COVID-19 therapy. *Advanced Drug Delivery Reviews*. 2020;167 Dec 100:1–17.
10. Huang Y, Tang Y, Lu R, Shi X, Song L, Wu X, et al. SARS-CoV-2 neutralization by monoclonal antibodies. *Cell Discov*. 2020 Jun 29;6(1):1–10.
11. Huang Y, Tang Y, Lu R, Shi X, Song L, Wu X, et al. Monoclonal antibodies for COVID-19 therapy and SARS-CoV-2 detection. *Journal of Cellular Biochemistry*. 2020 Jun 29;161(6):1047–56.